



CHAPTER Three

Data Communication Layers

Layered Tasks and Hierarchy

- Network communication is a complex process that involves multiple steps.
- To simplify this process, network models are divided into layers, each with specific tasks and responsibilities.

layered architecture, provides several benefits:

- **Modularity:** Each layer can be developed and modified independently, reducing complexity.
- **Flexibility:** New technologies or protocols can be introduced in one layer without affecting others.
- **Standardization:** Layered models facilitate the development of interoperable network devices and software.

Sender, Receiver, and The Carrier

- In any network communication, there are three essential components:
- **Sender:** The entity that initiates the communication by sending data.
- **Receiver:** The entity that receives the data sent by the sender.
- **Carrier:** The medium or channel that transports the data between the sender and receiver.

Rules & Network Protocols

- To ensure that data is transmitted and received correctly, network communication relies on a set of rules known as **protocols**.

These protocols define:

- **Data format:** The structure and encoding of data.
- **Timing:** When data is sent and received.

Rules & Network Protocols

- **Error detection and correction:**
Mechanisms for identifying and correcting errors in data transmission.
- **Flow control:** Mechanisms for regulating the flow of data to prevent congestion.
- **Standardization:** Protocols are standardized, which means they are agreed-upon and followed by different manufacturers.
- This ensures that devices from different vendors can communicate effectively.

Common Network Protocols:

- **TCP (Transmission Control Protocol):**
Provides reliable, ordered delivery of data.
- **IP (Internet Protocol):** Enables the routing of data packets across networks.
- **HTTP (Hypertext Transfer Protocol):**
Used for transferring data on the World Wide Web.
- **SMTP (Simple Mail Transfer Protocol) :**protocol for sending emails across the internet.

- **HTTPS (Hypertext Transfer Protocol Secure):** A secure version of HTTP that uses SSL/TLS encryption to protect data transmitted over the web.
- **SSH (Secure Shell):** Provides secure network access, remote command execution, and file transfer.
- **FTP (File Transfer Protocol):** Used for transferring files between computers over a network.

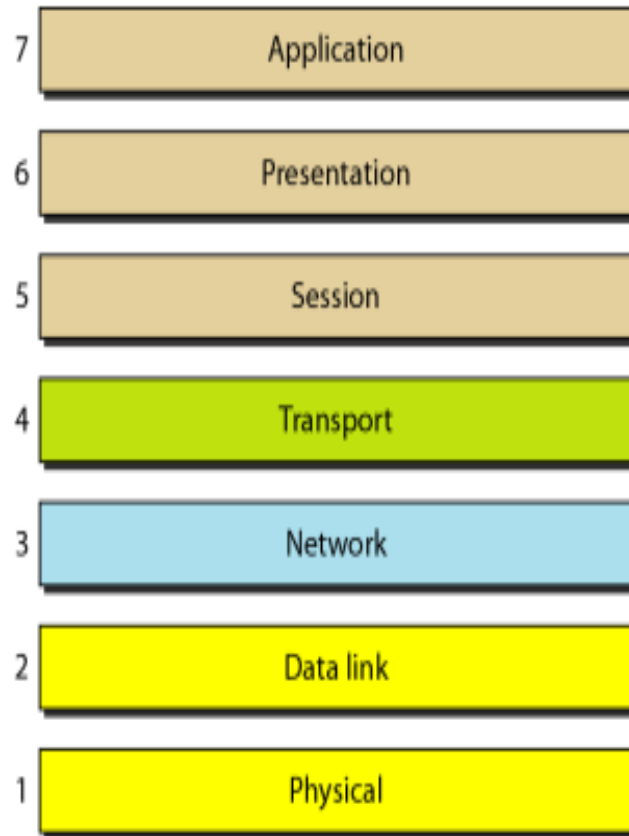
OSI reference model

- OSI stands for Open Systems Interconnection
- Created by International Standards Organization (ISO)
- Was created as a framework and reference model to explain how different networking technologies work together and interact

OSI reference model

- Communication between two devices is a complex process breaking into smaller discrete tasks improves interoperability and comprehension
- Each layer has specific functions it is responsible for
- All layers work together in the correct order to move data around a network

OSI reference model....



Top to bottom

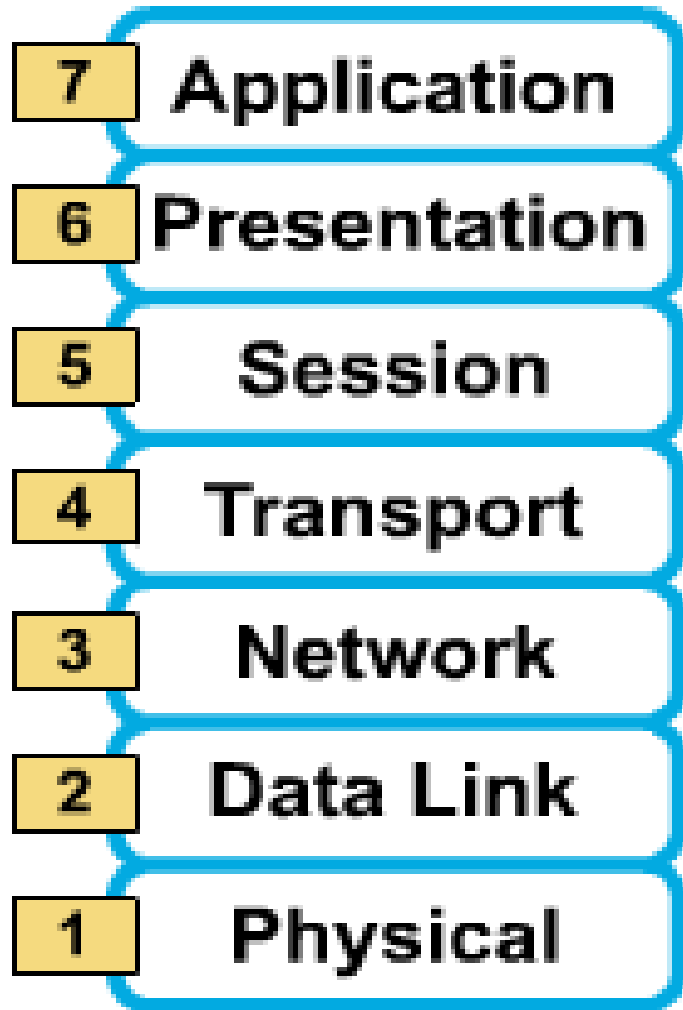
- **All People Seem To Need Data Processing**
- Bottom to top**
- **Please Do Not Throw Sausage Pizza Away**

OSI reference model....

- The **OSI** reference model is a layered, **abstract representation** created as a guideline for **network protocol design**.
- The OSI model divides the networking process into **seven logical layers**, each of which has unique functionality and to which are assigned specific services and protocols.

- 
- In this model, information is passed from one layer to the next, **starting at the Application layer** on the **transmitting host**, proceeding down the hierarchy to the **Physical layer**, then passing over the communications channel to the **destination host**;
 - where the information proceeds back up the hierarchy, **ending at the Application layer**.

The seven layers of OSI model



- *Data flow is divided into **seven smaller** and manageable steps.*
- *What devices operate at each layer?*
- *As a result, you will understand how to trouble shoot network problems.*

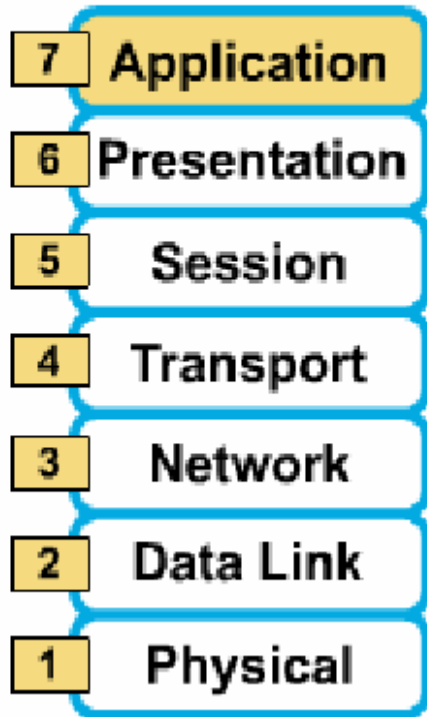


Functions of each layer

Application Layer

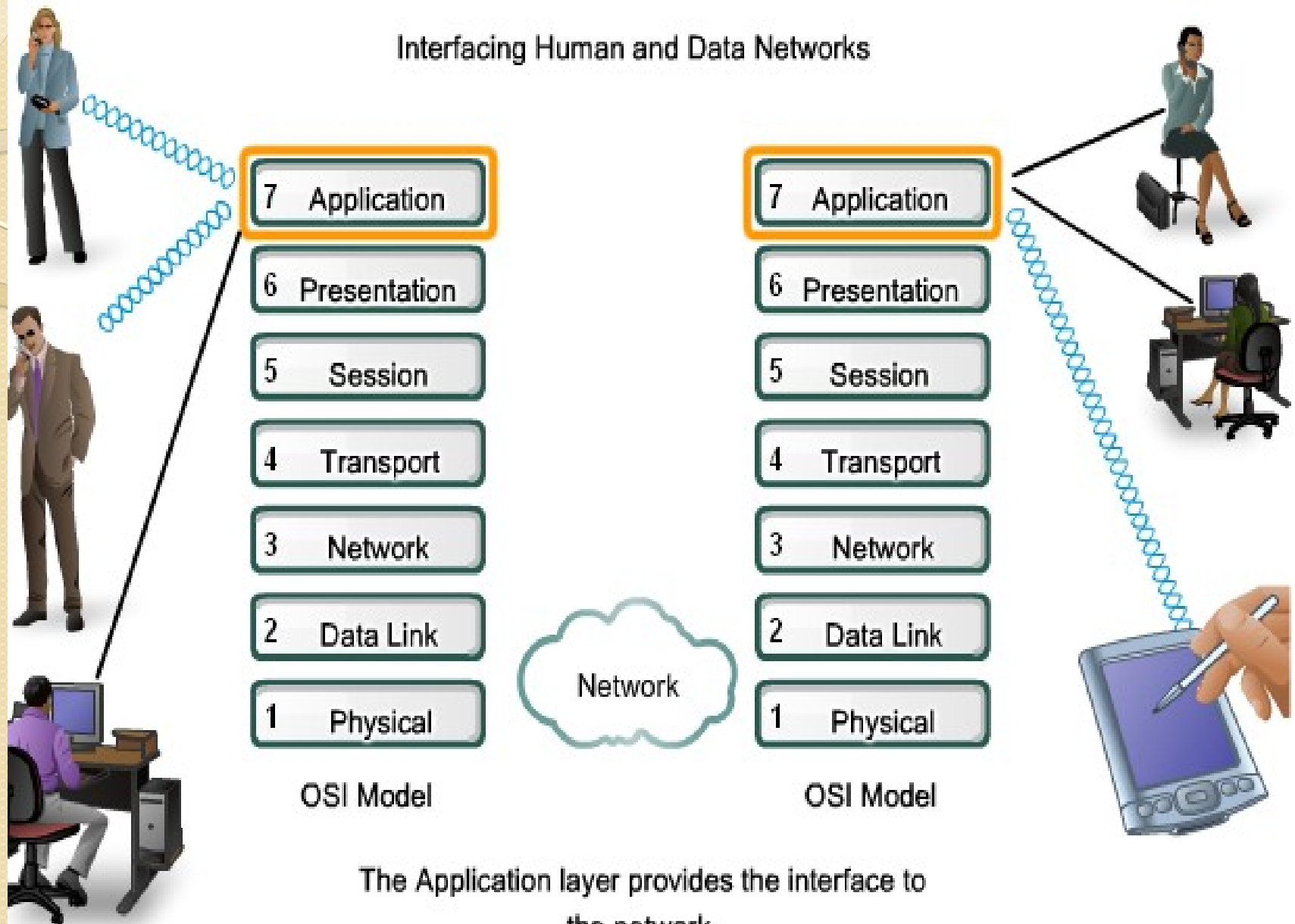
- ✓ The Application layer, Layer seven, is the top layer of both the **OSI** and **TCP/IP models**.
- ✓ It is the layer that provides the interface between the applications we use to communicate and the underlying network over which our messages are transmitted

:The application layer



- *Provides services to user's applications.*
- *Ensures the **availability of intended communication partner.***
- *Ensures the existence of sufficient network resources (ex. NIC).*
- ***Synchronizes and establishes agreement on error recovery.***
- *Data integrity.*

Interfacing Human and Data Networks



The Presentation layer

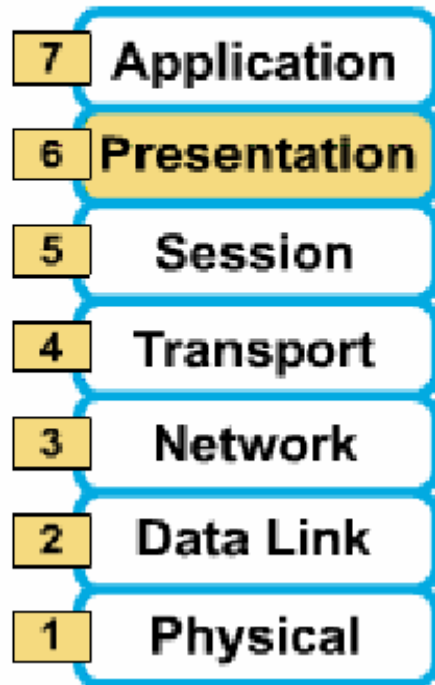
- It has three primary functions:

I. Coding and conversion of Application layer data to ensure that data from the source device can be interpreted by the appropriate application on the destination device.

II. Compression of the data in a manner that can be **decompressed** by the destination device.

III. Encryption of the data for transmission and the **decryption of data upon** receipt by the destination.

:The Presentation layer



• *services to user's applications.*

• *Data format.*

• *Insures data is readable by the receiving system.*

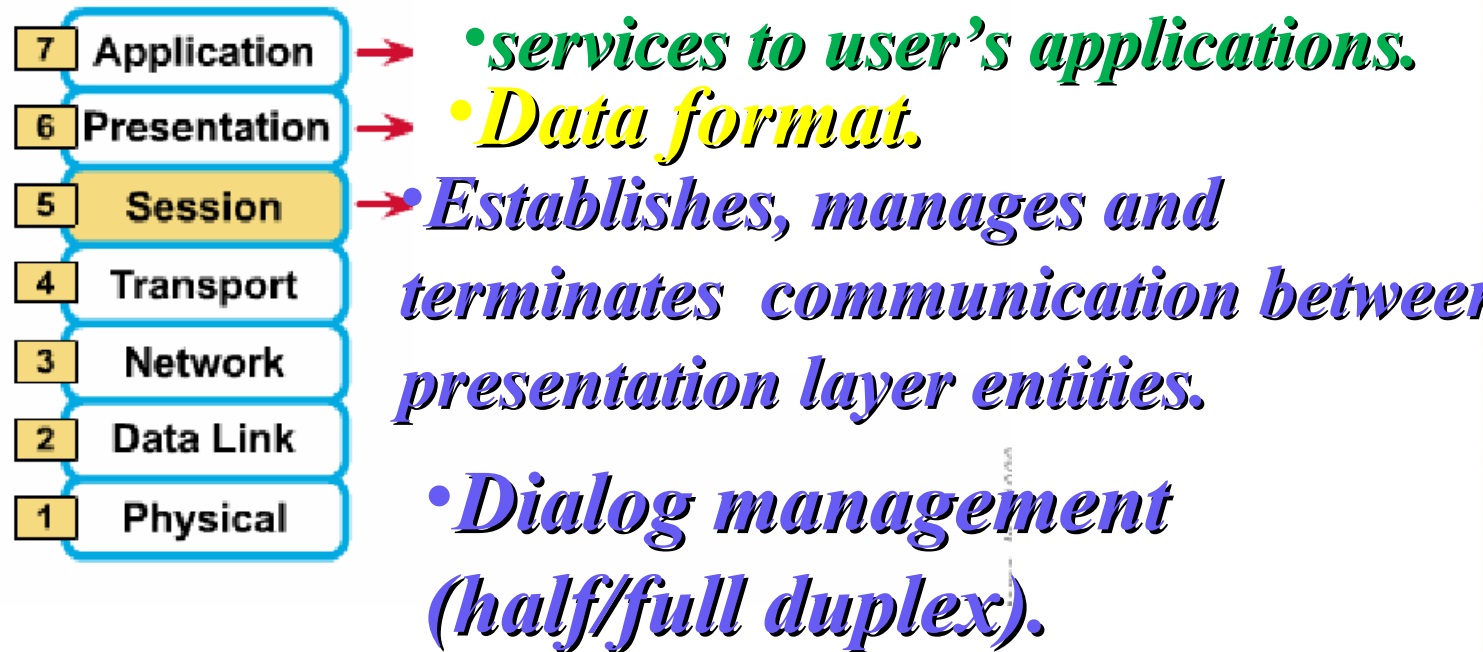
• *Translates between different data formats.*

• *Compression /decompression*

• *Encryption/ decryption.*

- 
- **Session Layer**
 - As the name of the Session layer implies, functions at this layer **create and maintain dialogs between source and destination applications.**
 - The Session layer handles the **exchange of information to initiate dialogs**, keep them active, and to restart sessions that are disrupted or idle for a long period of time.

:The session layer



Session layer cont...

- Provisions for efficient data transmission.
- *Reporting of session, presentation and application layer errors (errors above transport layer).*
- *Synchronization (checkpoints).*

Transport layer

- The Transport layer provides for the **segmentation of data** and the control necessary to reassemble these pieces into the various communication streams.
- Its primary responsibilities to accomplish this are:
 - Tracking the individual communication between applications on the source and destination hosts
 - **Segmenting data** and managing each piece
 - Reassembling the segments into streams of application data
 - Identifying the different applications

:The transport layer



• *Transport Layer Cont...*

- ✓ error recovery

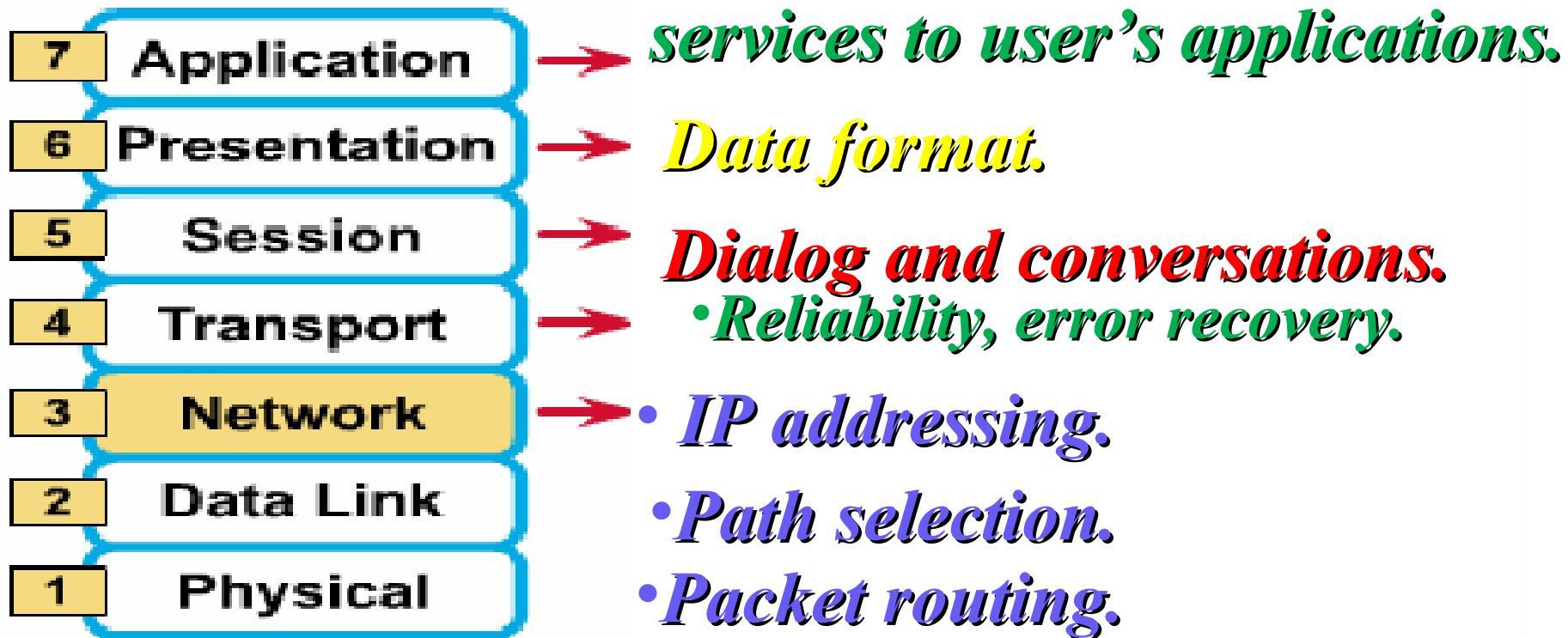
- ✓ *Flow control :*

- ✓ *Establishes , manages and terminates virtual circuits (not physical circuits).*

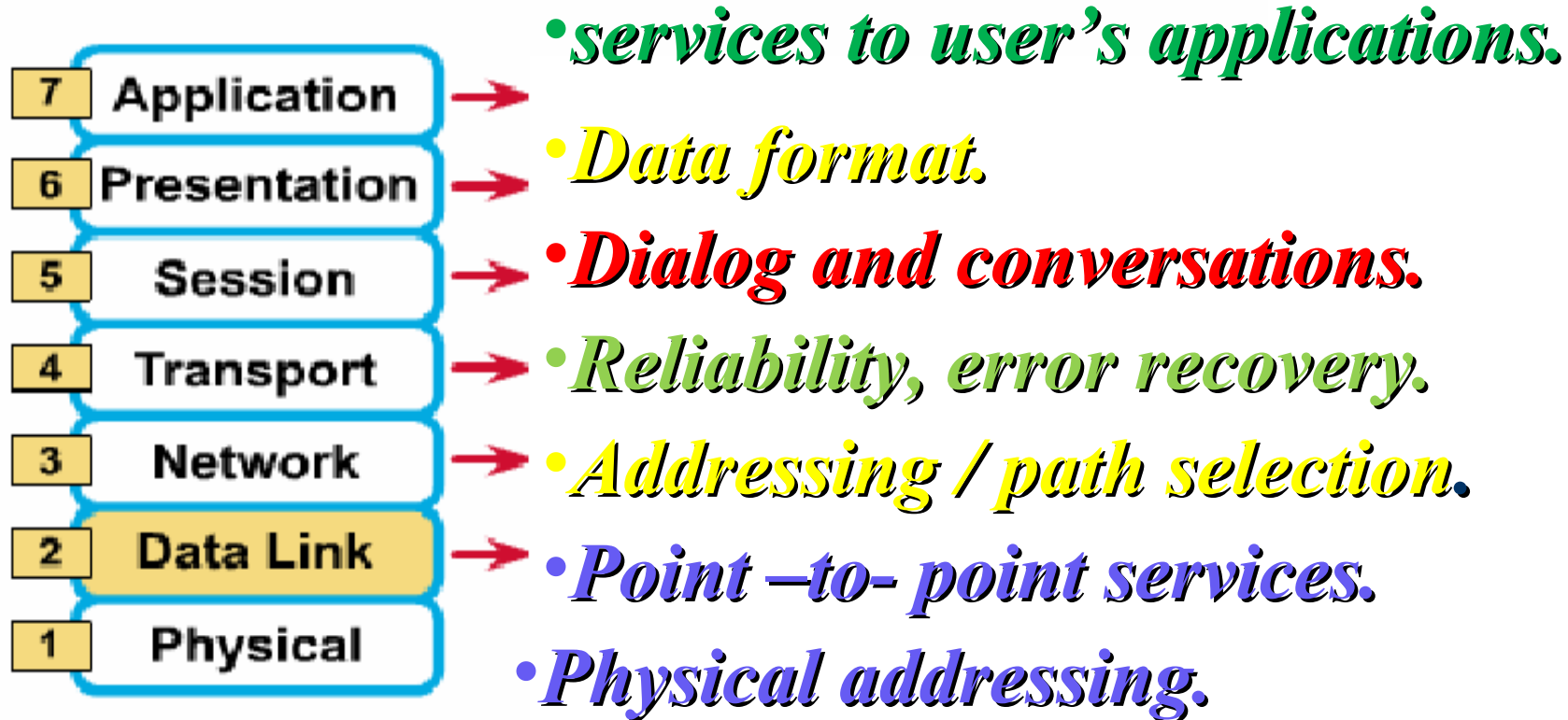
- ✓ *Shields the upper layers from transport details.*

- ✓ acknowledgment.

:The network layer



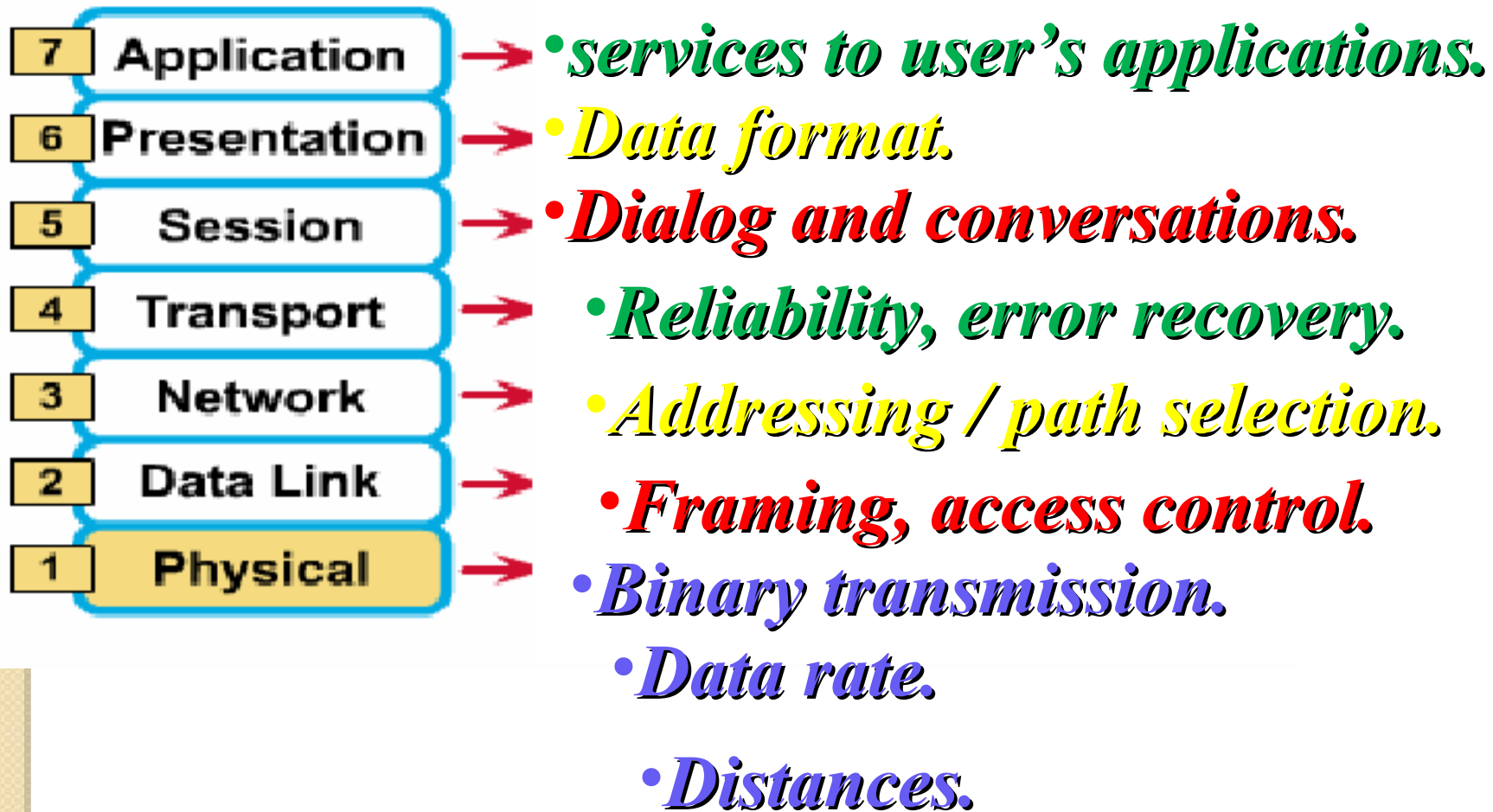
:The data link layer



Data link layer cont...

- ***Media access control.***
- ***Error detection.***
- ***Flow control.***
- ***Framing.***
- ***Reliability over the physical link.***

:The physical layer



Physical Layer Cont...

- ***Electrical characteristics of physical medium.***
- ***Pin assignment.***
- ***Timing of bits.***
- ***Voltage levels***

Data Flow and Encapsulation

- Data flow occurs when two devices are connected in a network with some kind of shared transmission medium.
 - 1) An application running on the source device creates some kind of data. This happens at the **Application layer**.
 - 2) The application needs to add encryption to that data. This will be done at the **Presentation layer**.
 - 3) At the Session layer it appends the **Session ID**.

Cont..

- 4) The **Transport layer** breaks the data into blocks of data which we call **Segments** . Every Segment also gets the **Port number** to identify which upper layer application needs to receive the data on the destination device
- 5) The **Network layer** takes the Segment, and appends the source and destination **IP address**. At that point the Segment becomes a **Packet**.

Cont..

6. 6) At Data Link layer the source and destination **MAC** address and the **CRC** is added. At this point we have a Frame.
7. The **Frame** then is sent to the **physical device** where it is translated into some kind of a signal (often call it **Bits**)

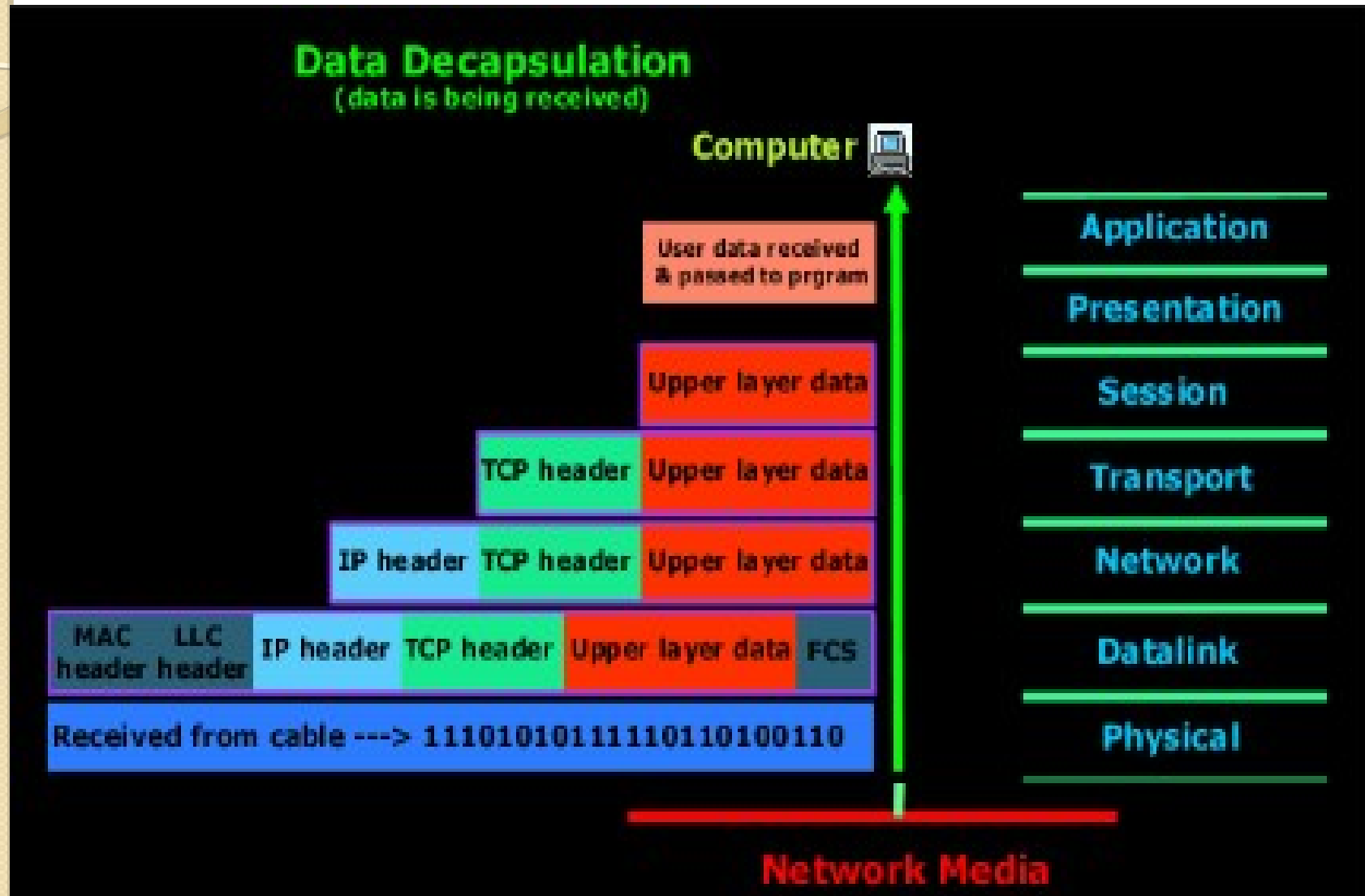
Decapsulation at the Destination Device

1. The destination device receives series of **bits** and interprets them as a Frame.
2. It removes MAC addresses and the CRC, and passes the data up to the Network layer.
3. IP addresses are removed and the Packet is forwarded up to the Transport layer .
4. The Port number is looked at and the Segment gets forwarded up the to the appropriate application specified by the Port number.
- 5.

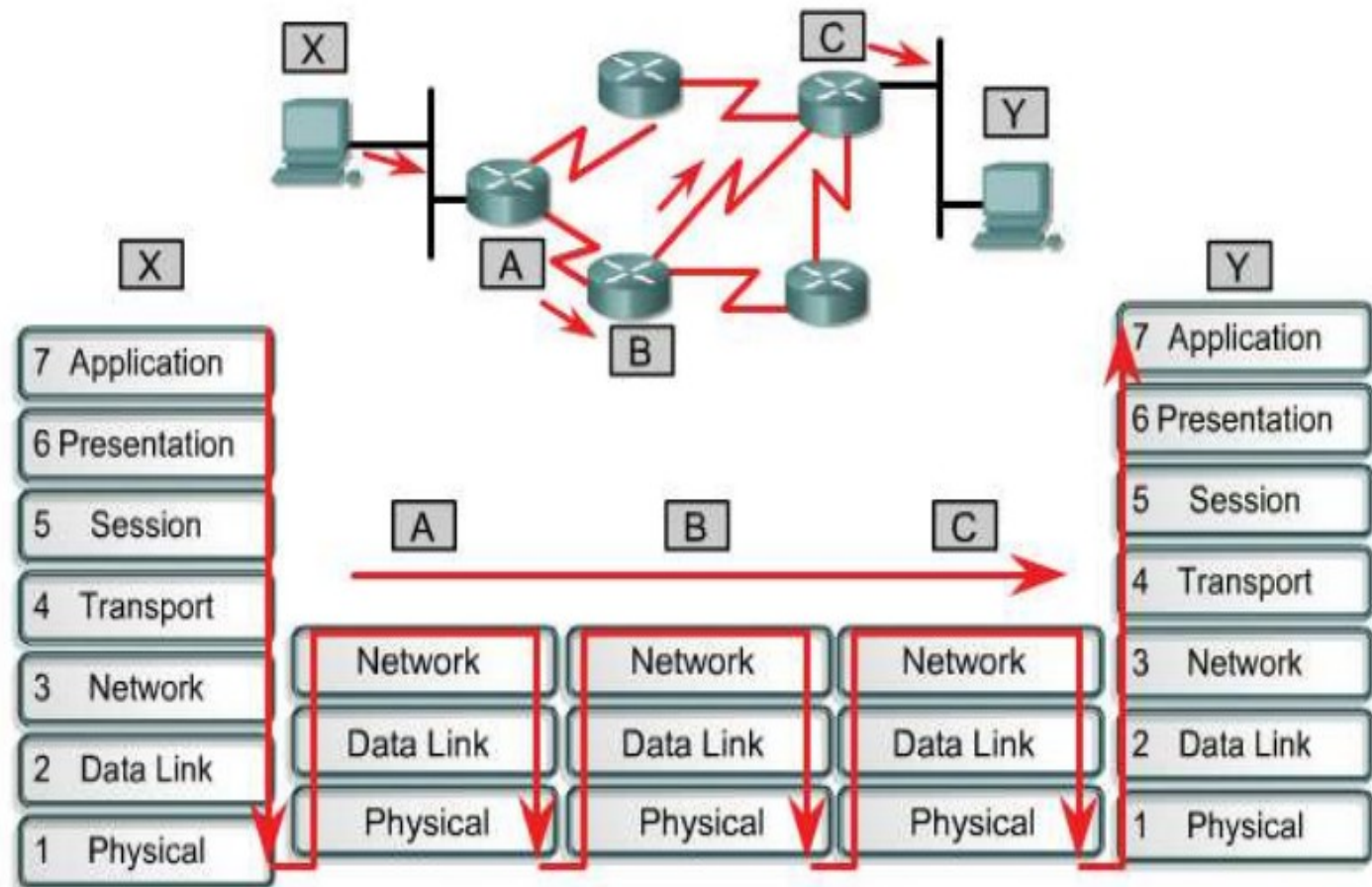
Cont..

5. The Session ID is used.
6. Encryption will be removed.
7. The data in its original form is presented to the application that needs to interpret it.

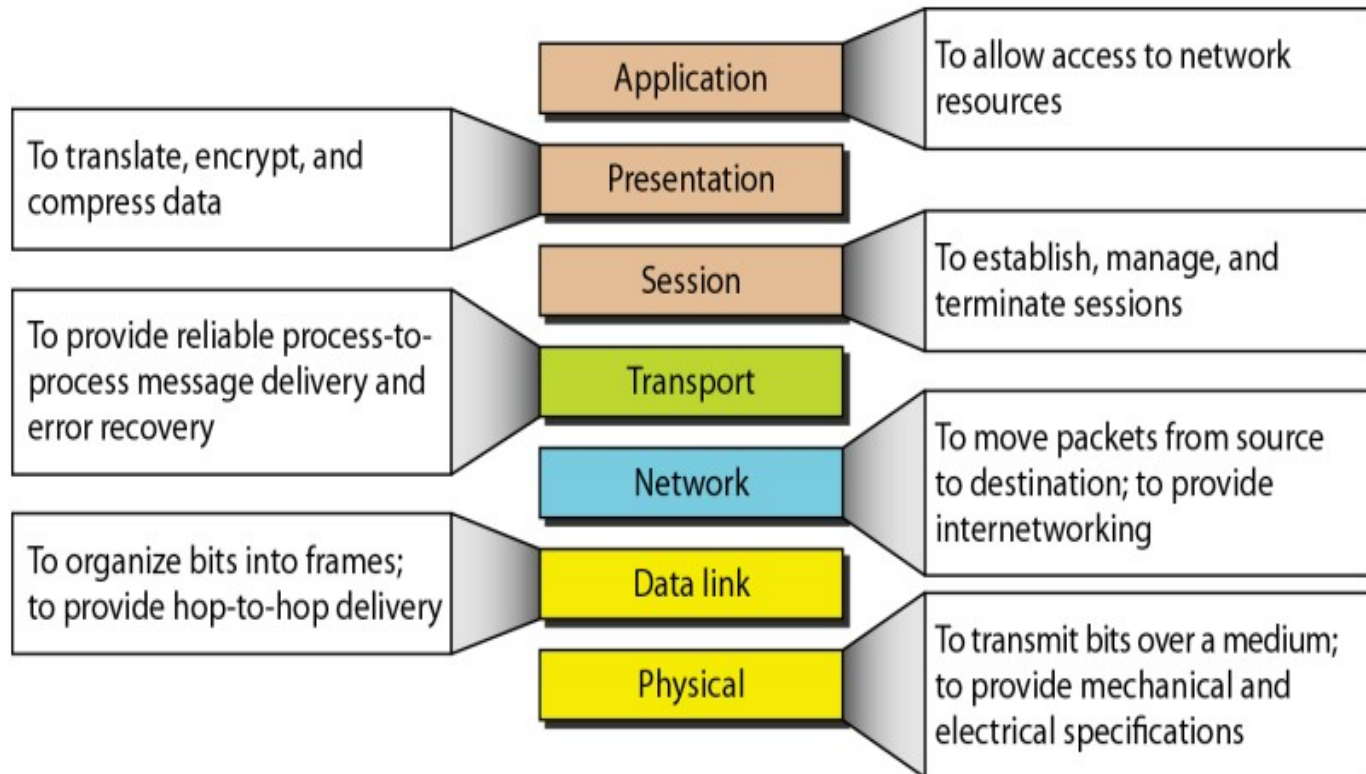
Data decapsulation



Example: Draw the data flow through a network ?



Summery of layers

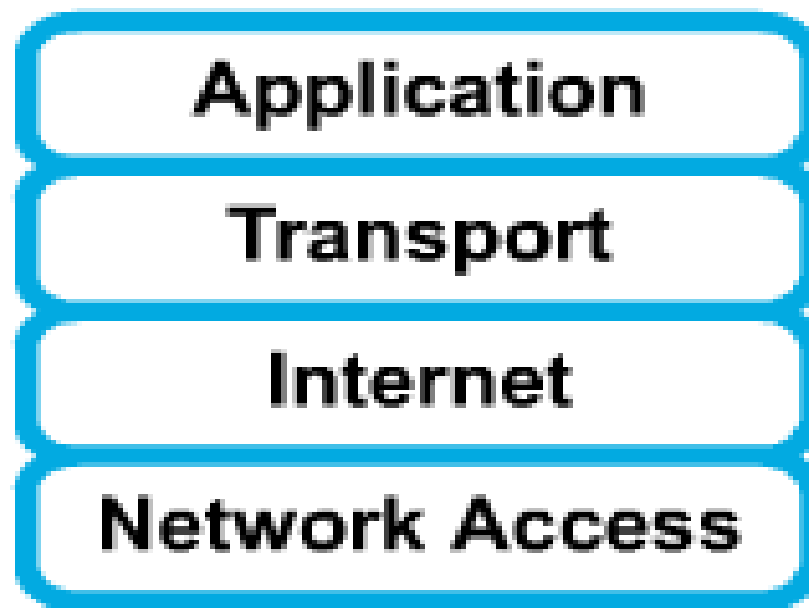




TCP / IP model

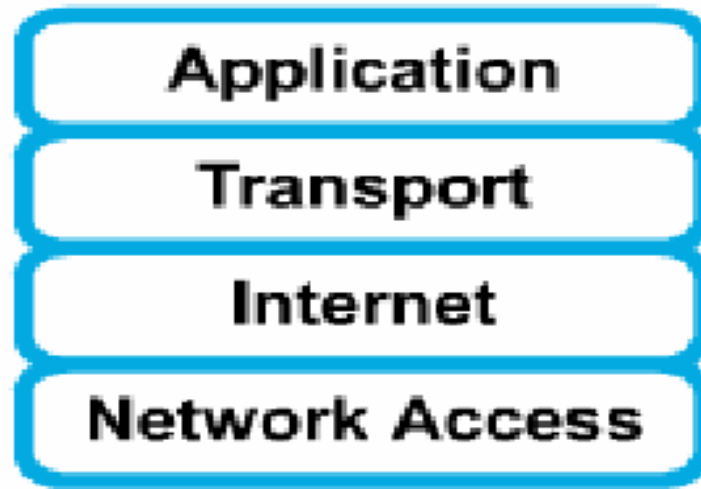
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- The U.S. Department of Defense (*DoD*) created the TCP/IP reference model because it wanted a network that could survive any conditions, even a nuclear war.
 - The main purpose was to move packets from any point to any other point regardless of the condition of any particular node.

The TCP/IP Model



:The application layer

- Session and Presentation layers are thin.
- TCP/IP model combines the functions of the upper three layers into **one layer**.



:The transport layer

- Deals with QoS issues (TCP protocol):
 - reliability:
 - ✓ Error recovery.
 - ✓ Acknowledgment.
 - flow control:
 - ✓ Sliding windows.
- TCP – connection oriented protocol.

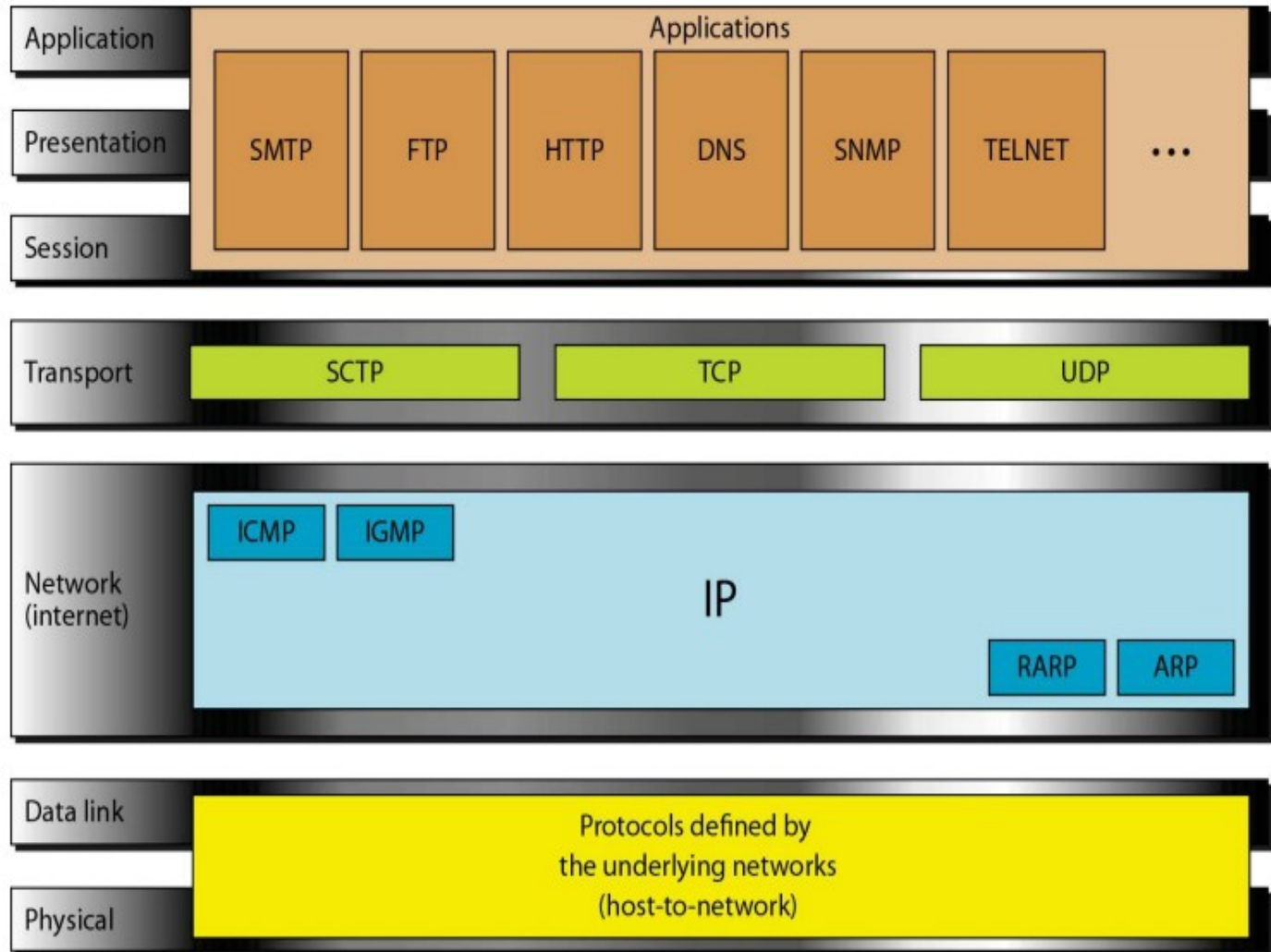
:The internet layer

- Path determination.
- Packet switching.
- IP addressing.

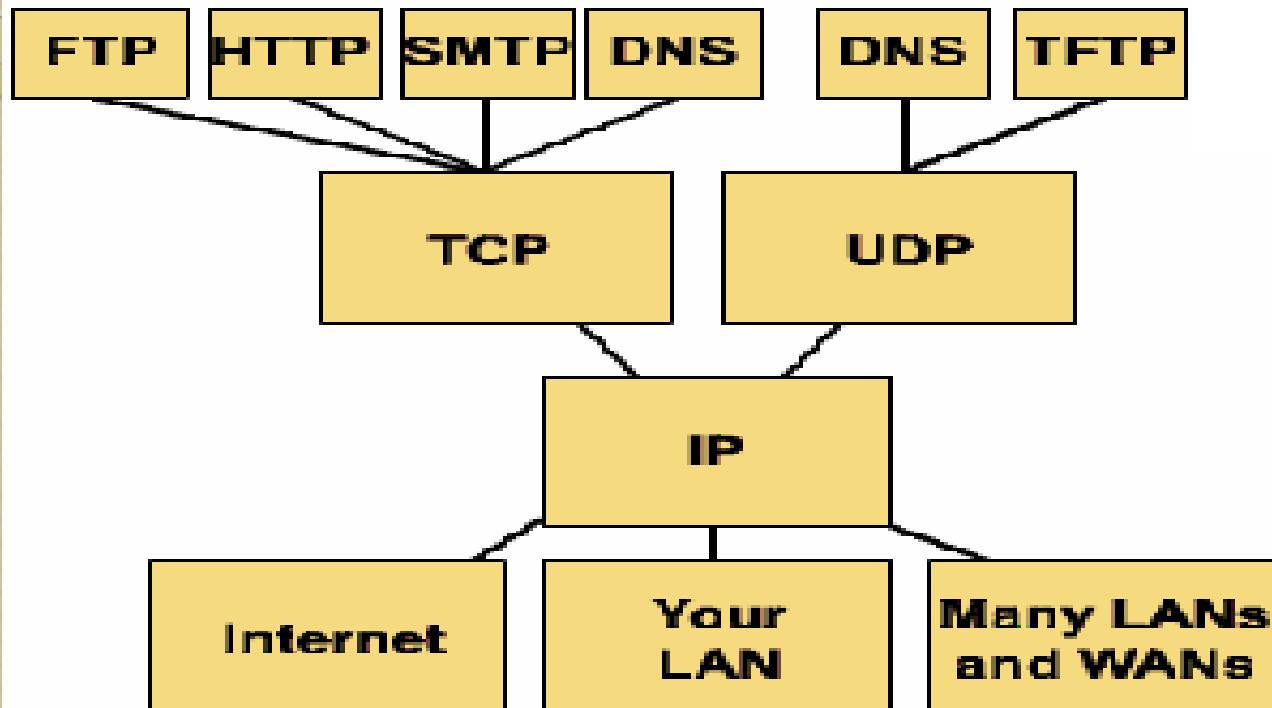
:The network-access layer

- Host-network layer.
- Combines the functions of physical and data link layers.
- It includes the WAN and LAN technology.

Cont.. *TCP/IP and OSI model*

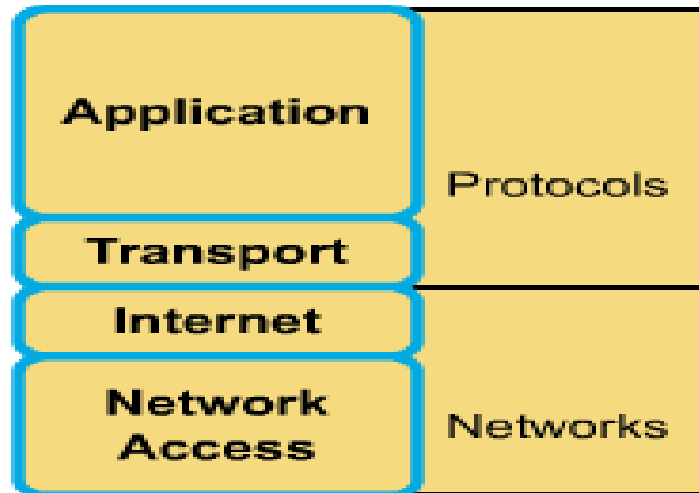


:TCP / IP protocol graph

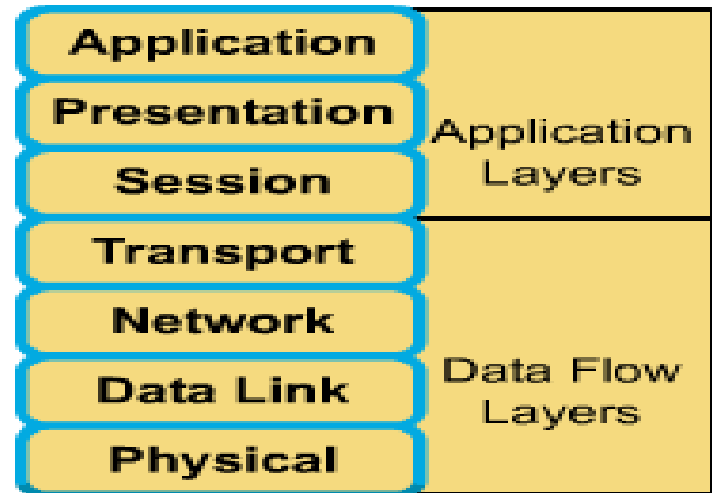


Comparing TCP/IP with OSI

TCP/IP Model



OSI Model



Similarities:

- ✓ both have layers
- ✓ both have application layers.
- ✓ both have comparable transport and network layers
- ✓ networking professionals need to know both

Differences

- ✓ TCP/IP combines the presentation and session layer issues into its application layer
- ✓ TCP/IP combines the OSI data link and physical layers into one layer(Host-network layer.)
- ✓ TCP/IP appears simpler because it has fewer layers
- ✓ TCP/IP protocols are the standards around which the Internet developed, so the TCP/IP model gains credibility just because of its protocols.

OSI #	OSI Layer Name	TCP/ IP#	TCP/IP Layer Name	Encapsul. Units .
7	Application	4	Application	Data
6	Presentation	4	Application	Data
5	Session	4	Application	Data
4	Transport	3	Transport	Segment
3	Network	2	Internet	Packet
2	Data link	1	Netw. access	Frame
1	Physical	1	Netw. access	Bits

Bits, Datagram, packet & Frames

Bits

- The smallest unit of data in computing.
- A bit can represent two states, typically represented as 0 or 1.
- **Analogy:** Imagine a light switch: it can be either "on" (1) or "off" (0).

Datagram

- A self-contained unit of data transmitted over a network.
- **Connectionless:** Datagrams are sent independently without establishing a prior connection.
- **Variable Size:** Datagrams can vary in size.
- **Routing:** Each datagram is routed independently through the network based on its destination address.
- **Analogy:** Imagine sending letters through the postal system. Each letter is a separate unit that travels independently to its destination.

Packet

- A unit of data transmitted over a network.
- **Key Characteristics:**
 - Can be connection-oriented or connectionless.
 - May belong to a larger stream of data.
 - Often has a fixed size or a maximum size limit.
- **Analogy:** Think of packets like individual boxes within a larger shipment. They may be part of the same shipment but are handled individually during transport.

Frames

- A unit of data transmitted over a physical link in a network.
- **Key Characteristics:**
 - **Includes header and trailer information:** The header contains information like the source and destination addresses, while the trailer may contain error-checking information.
 - **Specific to the physical layer:** Frames are directly related to the physical hardware and transmission medium.
- **Analogy:** Imagine frames as individual segments of a message being sent over a telephone line. Each segment is a complete unit with its own start and end markers.

In Summary:

- **Bits:** The fundamental building blocks of data.
- **Datagrams:** Independent units of data often used in connectionless networks (like the internet).
- **Packets:** Units of data that may or may not be part of a larger stream.
- **Frames:** Units of data transmitted over a physical link, containing information specific to the transmission medium.